DSA0405
Fundamentals of Data Science

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CO1 - AT1 Short Answers

1. What is Data science? Explain its primary objective.

Ans: Data science is the process of collecting, analyzing, and interpreting data using scientific methods, algorithms and technology. objective is to extract useful insights from data.

2. Why is Data science important in today's digital world? mention any four reasons.

Ans: Data science helps organizations make informed decision based on data instead of assumptions. It improves business efficiency; Predicts future trends, enhances customer experience, and supports automation through Artificial intelligence.

3. list any four benefits of Data science in business and society

Ans: Data science improves decision-making, increases business profits, detects fraud, and enhances healthcare services. It also helps governments, industries, and organizations solve real-world problems effectively.

4. Mention four real-world applications of Data science.

Ans: Data science is used in healthcare for disease prediction, banking for fraud detection, e-commerce for product recommendations, and transportation for traffic prediction. It is also widely used in education, agriculture and social media Analysis.

What are the different facets of data? Briefly explain each.

The main facets of data are structured, semi-structured, and unstructured data. structured data is organized in tables, semi structured data has partial organizations like XML/JSON and unstructured data includes images, videos, emails and text.

Differentiate b/w structured, semistructured and unstructured with examples

→ structured data is organized in rows and colums
Ex: Student database, Excel sheet.
→ Semi-structured data has hashtags or metadata but no fixed format
Ex: XML, JSON
→ unstructured data has no predefined format.
Ex: images, videos, audio files.

7. What is Big data? State its key characteristics (5Vs)

Ans: Big data refers to extremely large and complex datasets that cannot be processed using traditional tools.

- 1) Volume
2) Velocity
3) Variety
4) Veracity
5) Value

8. Explain the Big data ecosystem. Name any four technologies used in it

Ans: The big data ecosystem consists of tools used for collecting, storing, processing, analyzing and visualizing large amounts of data.
Some popular technologies are Hadoop, Spark, Hive and HBase

9. What are the major phases of the Data science process.

Ans: The Data science process includes data collection, data cleaning, data analysis, model building, model evaluation and deployment.
These phases help transform raw data into useful and predictions

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Ans: Data retrieval is the process of obtaining data from different storage systems for analysis. Common sources includes databases and web APIs

11. Why is data collection considered an important step in the Data science process?

Ans: Data collection provides the raw information needed for analysis and model building.
Accurate and relevant data improves the quality and reliability of result

12. What is data cleansing? list any four common data-cleaning techniques.

Ans: Data cleansing is the process of identifying and correcting errors in a dataset.
Common techniques include removing duplicates, handling missing values, correcting inconsistent data, and removing outliers.

13. what is data integration? why is it necessary in Data science?

Ans: Data integration is the process of combining data from multiple sources into a single dataset. It Provides a complete view of information and improves the accuracy of analysis.

14. Explain data transformation with suitable examples.

Ans: Data transformation is the process of converting data into a suitable format for analysis. Examples include converting text to numbers, normalizing values, and changing date formats

15. what is data analysis? mention any four techniques used for data analysis.

Ans: Data analysis is the process of examining data to discover useful Patterns and insights. common techniques include descriptive analysis, regression analysis, classification, and clustering.

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93%